Couldabeens v2: Player-Level Exit Hazard and Tax Pressure, 1985–2024

couldabeens243 v2 (Phase 3)

The unit of analysis is the qualifying player-season (G $\geq$ 30 position / IP $\geq$ 50 pitcher), 1985–2024: 30292 seasons, 6053 players, exit rate 7.6%. "Tax pressure" is the player's team payroll as a within-year percentile (splice- and proration-invariant), with the payroll/CBT-threshold ratio as a secondary measure from 2003. The couldabeen question, restated at the player level: **conditional on performance and age, did playing for an expensive roster raise the probability that a good player's season was his last, more so after the 2003 luxury tax?**

Model A: couldabeen-grade seasons (headline)

Seasons whose WAR clears that year's rookie-mean threshold – the players the 2020 report worried about losing:

Table 1: Exit hazard, couldabeen-grade seasons (n = 16735; player-clustered SEs)

	Estimate	Std. Error	z value	Pr(>|z|)
WAR	-0.862	0.082	-10.517	0.000
classpos	0.086	0.119	0.721	0.471
pay_pctile	-0.403	0.282	-1.431	0.152
post	-0.201	0.233	-0.864	0.388
pay_pctile:post	0.165	0.386	0.428	0.669

Payroll-percentile slope on the exit logit: pre-tax -0.403; post-tax -0.238 (SE 0.272, z = -0.87). A positive post-tax slope means good players on expensive rosters exit more.

Model B: over the tax line, 2003+

Table 2: Exit hazard, couldabeen-grade seasons 2003+ (n = 9149, 784 on over-threshold teams)

	Estimate	Std. Error	z value	Pr(>|z|)
WAR	-0.788	0.111	-7.088	0.000
over	-0.086	0.274	-0.315	0.753

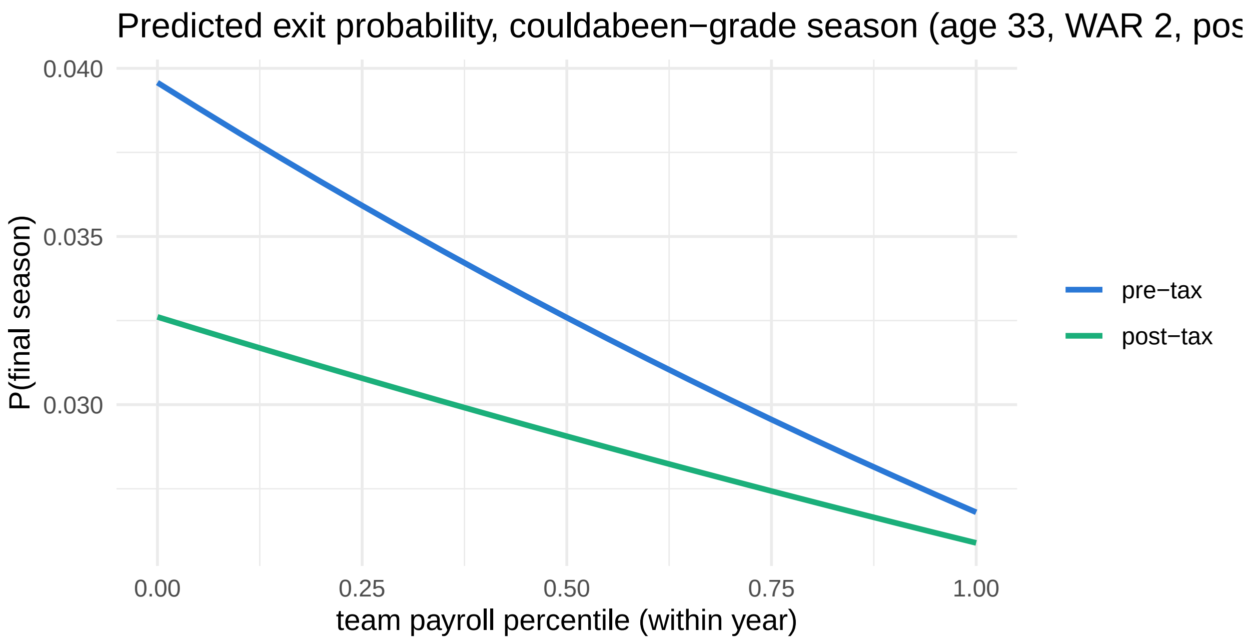
Model C: does the payroll effect depend on being good?

All qualifying seasons, three-way interaction (WAR x payroll percentile x post-tax era):

Table 3: All qualifying seasons (n = 29971)

	Estimate	Std. Error	z value	Pr(>|z|)
WAR	-0.850	0.064	-13.227	0.000
pay_pctile	-0.449	0.129	-3.489	0.000
post	0.065	0.093	0.691	0.490
WAR:pay_pctile	-0.014	0.111	-0.128	0.898
WAR:post	-0.125	0.090	-1.388	0.165
pay_pctile:post	0.017	0.167	0.103	0.918
WAR:pay_pctile:post	0.135	0.156	0.867	0.386

Predicted hazard



Verdict

Post-tax payroll-percentile effect on good players' exit hazard: z = -0.87. Over-the-threshold team effect (2003+): z = -0.32. Interpret positive, significant values as evidence for the tax-squeeze mechanism; values near zero say roster expense does not predict good players' exits once age and performance are controlled.